DESKTOP-Group TLP-GREEN

Or how creating your own CTI may lead to a cybercriminal's arrest

Tom Ueltschi
Swiss Post CERT
15. May 2024



The Story (of my life – well not the whole)

1. Short introduction
2. Prerequisites for "*Creating our own CTI*"
3. How we discovered "*DESKTOP-Group*" Threat Group
4. "*Sharing is caring*" ☺
5. Collaboration with INTERPOL
6. Is there a *Happy End*?

Short Introduction

C:> whoami /all

- Tom Ueltschi
- Swiss Post CERT / SOC / CSIRT since 2007 *(16 years!)*
- Focus & Interests: Malware Analysis, Threat Intel, Threat Hunting, Red / Purple Teaming, Detection Engineering
- Member of many trust groups & infosec communities
- FIRST SIG member (malware analysis, red teaming, CTI)
- Twitter: @c_APT_ure || @SwissPost_CERT

Swiss Post CERT
@SwissPost_CERT

CERT-Team @swisspost / @postschweiz
Cyber Defence for post.ch / AS12511
FIRST member @FIRSTdotOrg

(soon 17 years) ☺

Previous Presentations

- "Ponmocup Hunter" (Botnet malware)
  - SANS DFIR Summit 2013, DeepSec 2013, BotConf 2013, BotConf 2014
- "Threat Hunting with Sysmon Data" (and Splunk)
  - BotConf 2016, FIRST Con 2017, FIRST TC AMS 2018, BotConf 2018, CERT-EU Con 2019
- "DESKTOP-Group" – Tracking a persistent TG using email headers
  - BotConf 2019 (TLP-GREEN – not public)

All public slides linked on my blog:

http://c-apt-ure.blogspot.com/2017/12/is-this-blog-still-alive.html



Speaker history (about DESKTOP-Group)

- BotConf 2019 - DESKTOP-Group - Tracking a Persistent Threat Group (using Email Headers) [PDF] *(91 slides)*
- Reversing 2020 - Pushing the barriers of unique YARA uses [PDF] *(99 s.)*
- BotConf 2022 - LT: Advanced Persistent Speaker ☺ (DESKTOP-Group) [PDF]
- BotConf 2023 - LT: DESKTOP-Group or OPERA1ER [PDF]

Presentations available here: c-apt-ure.blogspot.com
http://c-apt-ure.blogspot.com/2022/01/who-is-desktop-group.html
https://github.com/c-APT-ure/my-public-stuff/

Prerequisites for Creating our own CTI

Make your Preventions work (for you)

- Restrict email attachments
  - Scan attachments with AV, sandboxing, (custom) YARA rules
  - Block executables, scripts, potentially unsafe file formats *(macros, .one)*
  - Based on file-extensions and/or file-types (magic bytes)
  - Manage exceptions / whitelists
  - Send blocked email to quarantine → your best honeypot & CTI source (?)
- Restrict web downloads
  - Same as with email attachments – as much as possible – though it's harder
  - Allow blocking on proxy for: *(incl. substring or regex)*
    - User-Agents, URL-patterns, TLD's
    - Allowing for exceptions

Remember 2015 BotConf Lightning Talks?

Creating your own
CTI in 3 minutes

TLP-AMBER

Tom Ueltschi / BotConf 2015
tueltschi@people.ops-trust.net / @c_APT_ure

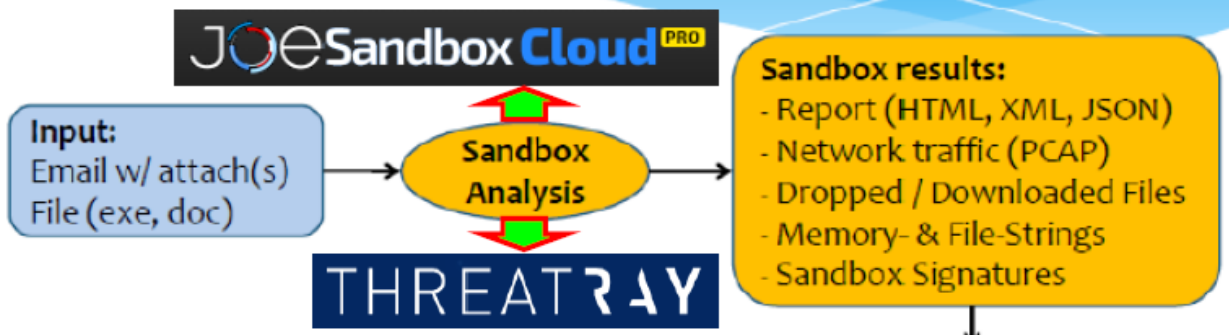
Automated Sandbox malware analysis

«Swiss Army Knive of Sandboxes» (commercial, private cloud)
http://www.joesecurity.org/joe-sandbox-technology [Blog]

Automation / Scripting:
- Extract mail attachs
- Upload samples to sandbox
- Download analysis results
  - report HTML/XML, PCAP, dropped files, file-/mem-strings
- Post processing
  - PCAP & tshark (DNS, HTTP, TCP)
  - XML & xquilla (files/reg keys created, mutexes, SB-sigs)
  - YARA scans of files, mem-strings & PCAP
  - VT hash lookups (submit sample & dropped files)

Tom Ueltschi / BotConf 2015

Automating Malware Analysis



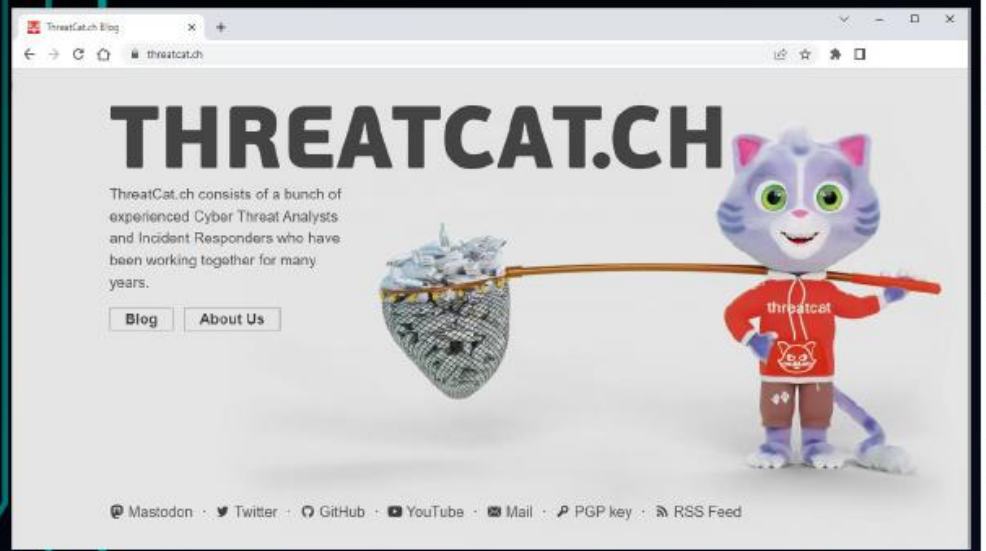
Post Processing
- XML Report & xquilla & xpath → files-, reg keys-, mutexes-, proc's created
- PCAP & tshark → DNS-, HTTP-requests, TCP connections (non-std ports)
- **Yara** rules & Files, PCAP, mem strings → File / Memory / Network patterns
- **VirusTotal** Filehash lookups, sample submits (optional) → AV detections

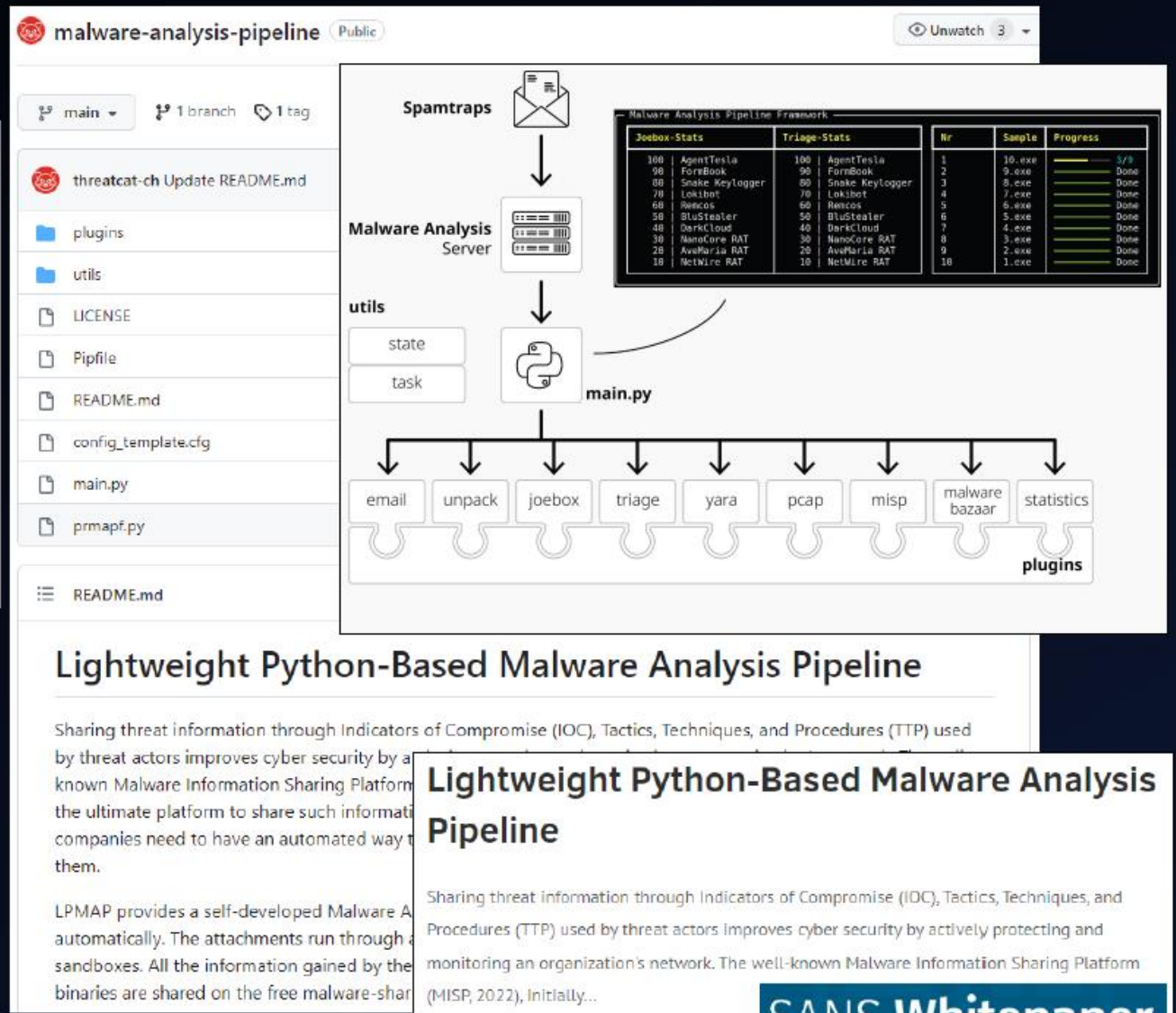
Behavior Analysis (Proc's, Files, Reg keys, Network, Persistence)

Sysmon Data in SIEM (Splunk)
Behaviors (Proc's) → Search Queries → Alerts & Hunting

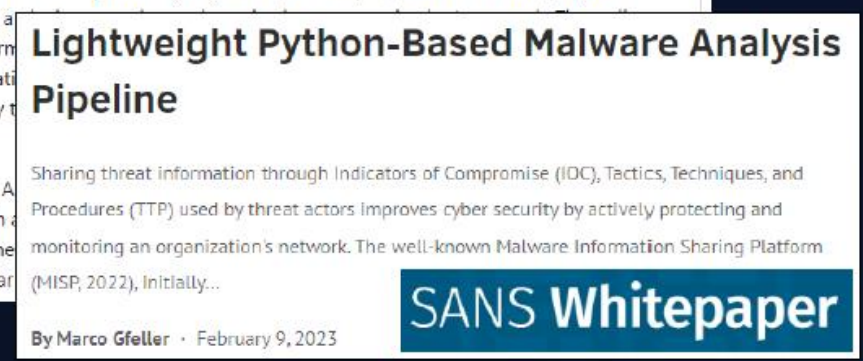
Botconf 2016 | Advanced Incident Detection and Threat Hunting using Sysmon and Splunk | Tom Ueltschi | TLP-WHITE Seite 49

ThreatCat MAPF









Sources:
https://threatcat.ch/
https://github.com/threatcat-ch/malware-analysis-pipeline
https://www.sans.org/white-papers/lightweight-python-based-malware-analysis-pipeline/

Does «size» really matter?
(Semi-)Automating Malware Analysis

- Number of analyzed malware samples
  - Per month → 50 to 400 (average ~230)
  - Per year → ~2'000 to ~3'500

```
2014 -> 1893
2015 -> 3184
2016 -> 3461
2017 -> 2409
2018 -> 1982
2019 -> 1997 (*)
```

→ **«Small numbers», but high value!**

Automated Sandbox malware analysis

Year	2013		2014		2015	
			134	2014-01	252	2015-01
			191	2014-02	261	2015-02
			290	2014-03	356	2015-03
			228	2014-04	251	2015-04
			137	2014-05	258	2015-05
			41	2014-06	320	2015-06
			81	2014-07	184	2015-07
	16	2013-08	146	2014-08	207	2015-08
	39	2013-09	134	2014-09	220	2015-09
	66	2013-10	206	2014-10	274	2015-10
	60	2013-11	175	2014-11	227	2015-11
	109	2013-12	130	2014-12		
Total	**290**		**1893**		**2810**	**4993**
Average	**58**		**158**		**255**	

Tom Ueltschi / BotConf 2015

Malware Analysis Statistics

	A	B	C	D
1	**Year**	**Count**	**Avg/mon**	**Avg/day**
2	2014	712	59	2.0
3	2015	3185	265	8.7
4	2016	3466	289	9.5
5	2017	2409	201	6.6
6	2018	1982	165	5.4
7	2019	2273	189	6.2
8	2020	3101	258	8.5
9	2021	3258	272	8.9
10	2022	3143	262	8.6
11	2023	2285	190	6.3
12	2024	550	46	1.5
13		**26364**		

Discovering DESKTOP-Group

Received header hostname = Message-ID host

```
Received: from relay12.mail.gandi.net ([217.70.178.232])
  by [REDACTED] with ESMTP/TLS/DHE-RSA-AES256-GCM-SHA384; 05 Apr 2019 18:13:44 +0200
Received: from DESKTOPHUHM1TV (unknown [154.0.26.84])
  (Authenticated sender: accounts@maslowgroup.net)
  by relay12.mail.gandi.net (Postfix) with ESMTPSA id 45986200014
  for <[REDACTED]>; Fri,  5 Apr 2019 16:13:19 +0000 (UTC)
MIME-Version: 1.0
From: "UPAEP - Clearing House" <Clearing.House@upaep.int>
Reply-To: Clearimg.House@upaep.int
To: "anne-claude.KELLY@upaep.int; alvaro.psetizki@upu.int" <[REDACTED]>
Subject: =?windows-1252?Q?Important:_Mise_=E0_jour_r=E9pertoire_IFS_/_Upda?=
  =?windows-1252?Q?te_directory_IFS?=
Content-Type: multipart/mixed;
  boundary="----=_NextPart_001_6EAE_5D306667.4C3E5736"
X-Mailer: Smart_Send_4_1_13
Date: Fri, 5 Apr 2019 18:13:15 +0200
Message-ID: <613649150232832260314[illegible]@DESKTOP-HUHM1TV>

(decoded) Subject: Important: Mise à jour répertoire IFS / Update directory IFS
```

DESKTOPHUHM1TV

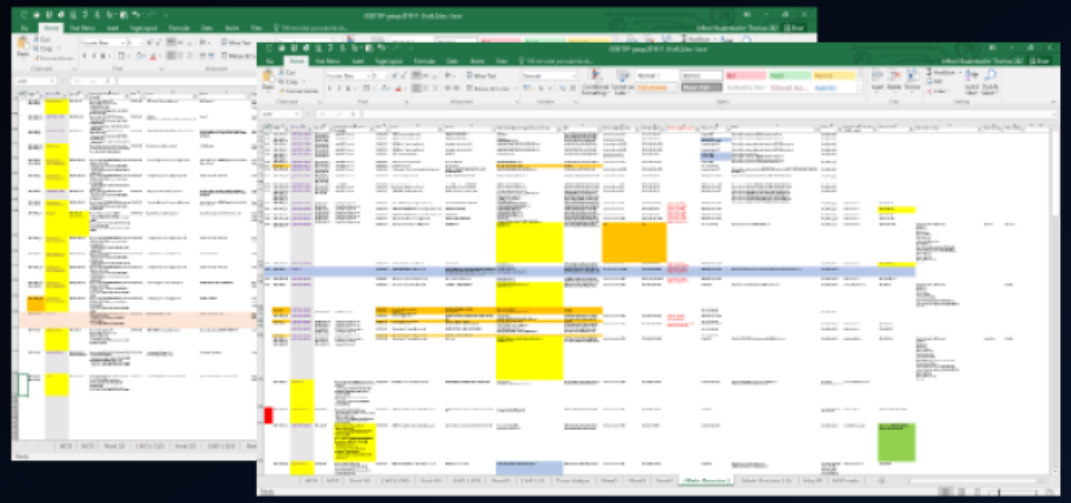
@DESKTOP-HUHM1TV>

Message-ID / (9) Desktop-/ (2) Server-names

Message-ID	Date from	Date to	Days	count
@DESKTOP-0LDSDAH	2018-02-08	2018-02-16	9	4
@DESKTOP-T4UN9D6	2018-02-12	2018-06-16	125	7
@DESKTOP-CBQP7F3	2018-02-22	2018-02-22	1	1
@DESKTOP-BHMUG0K	2018-07-04	2018-09-26	85	5
@DESKTOP-HUHM1TV	2018-07-21	2019-04-05	259	12
@DESKTOP-DDC429B	2019-03-31	2019-03-31	1	1
@DESKTOP-7U3H8EU	2019-05-11	2019-08-12	94	6
@DESKTOP-61D188I	2019-07-18	2019-09-25	70	7
@DESKTOP-FK2FFAC	2019-10-14	2019-10-17	4	3
WIN-P9NRMH5G6M8	2019-10-16	2019-10-24	9	6
WIN-N4R7BBAH231	2019-11-06	2019-11-06	1	1

Message-ID	Client IP	Date from	Date to	Days	count
@mailedge01	154.0.26.48 (annette.moukodi)	2019-05-28	2019-05-28	1	1
@mailedge01	185.247.228.17 (maloum.aboubakar)	2019-07-05	2019-07-11	7	3
@mailedge01	154.0.26.55 (fofou.sonfack)	2019-07-25	2019-10-22	90	3

First Hand Knowledge
Analyzing mail headers



- Date
- From (display-name / email)
- Subject
- Attachment(s) – Filename(s) / MD5 hash(es) → Malware Analysis
- Message-ID → Malware / RAT Family
- X-Mailer / User-Agent → C2 domain / IP / port
- X-Source-Auth / X-Sender / Authenticated-Sender
- X-Source-IP / X-Originating-IP
- Received headers → Client IP

Malware / RAT Families seen
sorted by chronological order

Malware family	Date from	Date to	Days	Count
NanoCore RAT	2018-02-08	2019-10-22	622	23
VBS/Dunihi / Houdini RAT	2018-02-11	2018-12-14	307	13
Imminent Monitor RAT	2018-02-12	2019-08-19	554	8
CyberGate RAT	2018-02-13	2018-02-13	1	1
Netwire RAT	2018-08-20	2019-05-28	282	5
WSH-RAT	2019-05-11	2019-11-12	186	6
Adwind RAT	2019-10-14	2019-11-06	24	11
Quaverse RAT / qRAT	2019-10-31	2019-10-31	1	1
Ave Maria RAT	2019-11-05	2019-11-05	1	1

- 1 CyberGate RAT sample confirmed
- 1 Quaverse RAT sample not [strongly] linked to this group

Expanding the Knowledge
Passive DNS for C2 domains & IPs

C2 Domains	resolved IPs	First seen	Last seen	0.0.0.0 since	sinkholed since	Days
ghanaandco.sytes.net	17	17.06.2017 00:00	28.05.2019 18:15	02.08.2019 15:29		711
goodattack.ddns.net	119	10.07.2017 01:52	18.02.2019 09:19	21.02.2019 19:00		588
etoilefiilante.ddns.net	59	14.07.2017 00:00	20.06.2018 14:23	05.07.2018 14:20		342
testultra.ddns.net	68	10.07.2018 09:21	02.07.2019 05:45	14.07.2019 10:13		357
chance2019.ddns.net	131	01.04.2019 00:00	13.10.2019 21:33	(still active)		196
worldcoupe.ddns.net	39	02.05.2019 00:00	04.07.2019 00:00	11.07.2019 20:26		63
etoilefiilante.duckdns.org	2	09.07.2019 00:00	16.07.2019 00:00		16.07.2019 00:00	7
etoilefiiilante.duckdns.org	2	17.07.2019 00:00	04.09.2019 09:54		05.09.2019 00:00	49
deaphnote.ddns.net	12	15.07.2019 00:00 09.09.2019 00:00	13.10.2019 19:26	(still active)		35
audreysaradin.no-ip.org	199	09.04.2015 00:00	05.11.2019 16:00	22.11.2019 06:23		1672
richordie.sytes.net	5	31.10.2019 00:00	13.11.2019 10:04	20.11.2019 05:17		13
Total	653					

- Longest used domain: 1’672 days (since April 2015)
- 0.0.0.0 = «inactive» (?) // 192.169.69.25 = sinkhole.hyas.com

Sharing is caring ☺

How it started… (Feb – Mar 2018)

Filters: Tag: DESKTOP-group ✕

Filters: Tag: DESKTOP-group | My Events | Org Events

Published	Creator org	Owner org	ID	Clusters	Tags	#Attr.	Creator user	Date	Info	Distribution
✓	SwissPost	SwissPost	264		DESKTOP-group, NanoCore RAT, tlp:green	27	thomas.ueltschi@post.ch	2018-02-08	servicecourrier@bcao.com / "INFORMATION URGENT !!" / BCAO.wsf	All
✓	SwissPost	SwissPost	265		DESKTOP-group, NanoCore RAT, VBS/Dunihi / H-Worm RAT, tlp:green	35	thomas.ueltschi@post.ch	2018-02-11	servicecourrier@bcao.com / "INFORMATION URGENT !!" / BCAO.vbs	All
✓	SwissPost	SwissPost	267		DESKTOP-group, Imminent Monitor RAT, tlp:green	22	thomas.ueltschi@post.ch	2018-02-12	marie.fourny@upu.int / "Clearing Module Automatisation IFS" / "MODULE AUTOMATISATION IFS CLEARING.wsf"	All
✓	SwissPost	SwissPost	269		DESKTOP-group, Imminent Monitor RAT, tlp:green	22	thomas.ueltschi@post.ch	2018-02-12	marie.fourny@upu.int / "Clearing Module Automatisation IFS" / "MODULE AUTOMATISATION IFS CLEARING.hta"	All
✓	SwissPost	SwissPost	266		DESKTOP-group, NanoCore RAT, CyberGate RAT, tlp:green	42	thomas.ueltschi@post.ch	2018-02-13	servicecourrier@dsi.com / "URGENT !!!" / "Notre De Service.hta"	All
✓	SwissPost	SwissPost	268		DESKTOP-group, tlp:green	14	thomas.ueltschi@post.ch	2018-02-16	d.courrier@poste.tn / "Information Urgent !!" / "NOTRE DE SERVICE.pdf.hta"	All
✓	SwissPost	SwissPost	271		DESKTOP-group, tlp:green	7	thomas.ueltschi@post.ch	2018-02-22	Direction@central.com / "Alerte de sécurité"	All
✓	SwissPost	SwissPost	273		DESKTOP-group, Imminent Monitor RAT, tlp:green	17	thomas.ueltschi@post.ch	2018-03-02	khaled.zargelayoun@monetiquetunisie.com / "URGENT : TRANSACTIONS SUSPECTE - SWIFT TRANSFERS MT103" / "SWIFT TRANSFERS MT103-MT300.jpeg.scr"	All
✗	SwissPost	SwissPost	274		DESKTOP-group, VBS/Dunihi / H-Worm RAT, tlp:green	19	thomas.ueltschi@post.ch	2018-03-08	info@france24.fr / "URGENT: NEYMAR QUITTE LE PSG" / https://void[.]cat/d69a641b797589eb2bc7850403b4688f57482ef6	All
✓	SwissPost	SwissPost	270		DESKTOP-group, VBS/Dunihi / H-Worm RAT, tlp:green	18	thomas.ueltschi@post.ch	2018-03-09	info@accescanada.com / "Urgent: Clôture du dossier OG009/2FP" / https://void[.]cat/d69a641b797589eb2bc7850403b4688f57482ef6	All
✓	SwissPost	SwissPost	272		DESKTOP-group, VBS/Dunihi / H-Worm RAT, tlp:green	47	thomas.ueltschi@post.ch	2018-03-19	info@euronews.com / "URGENT: La Russie met en garde la France"	All

DESKTOP-Group research - correlation event (work in progress...)

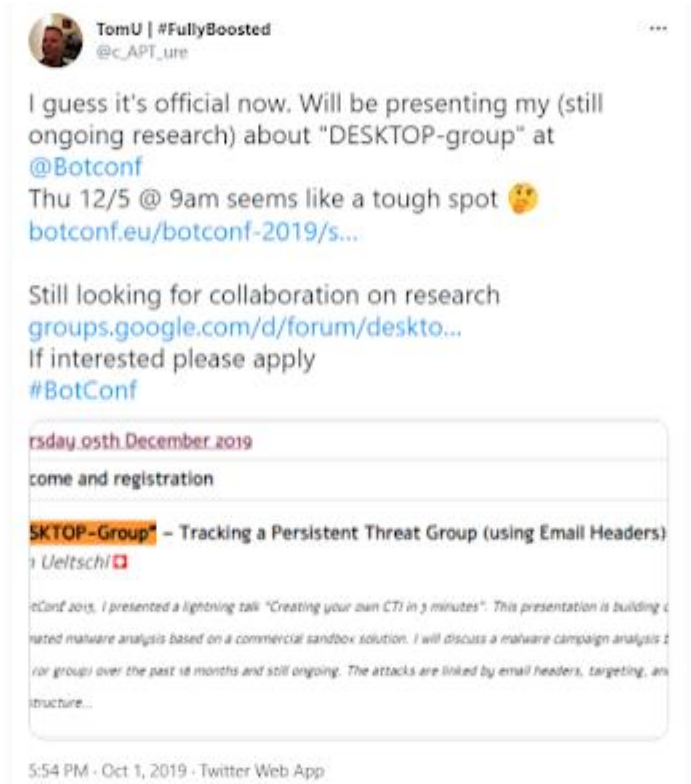
Event ID	5869
UUID	d779bc29-0c4b-4a0f-8886-b486e7c6d668
Creator org	SwissPost
Owner org	SwissPost
Creator user	thomas.ueltschi@post.ch
Tags	DESKTOP-group x tlp:green x
Date	2023-12-22
Threat Level	Medium
Analysis	Ongoing
Distribution	All communities
Info	DESKTOP-Group research - correlation event (work in progress...)
Published	Yes (2024-04-15 21:03:33)
#Attributes	1162 (0 Objects)
First recorded change	2023-12-22 09:34:00
Last change	2024-03-20 20:06:29

#Attributes	1162 (0 Objects)
First recorded change	2023-12-22 09:34:00
Last change	2024-03-20 20:06:29

Wednesday, January 26, 2022

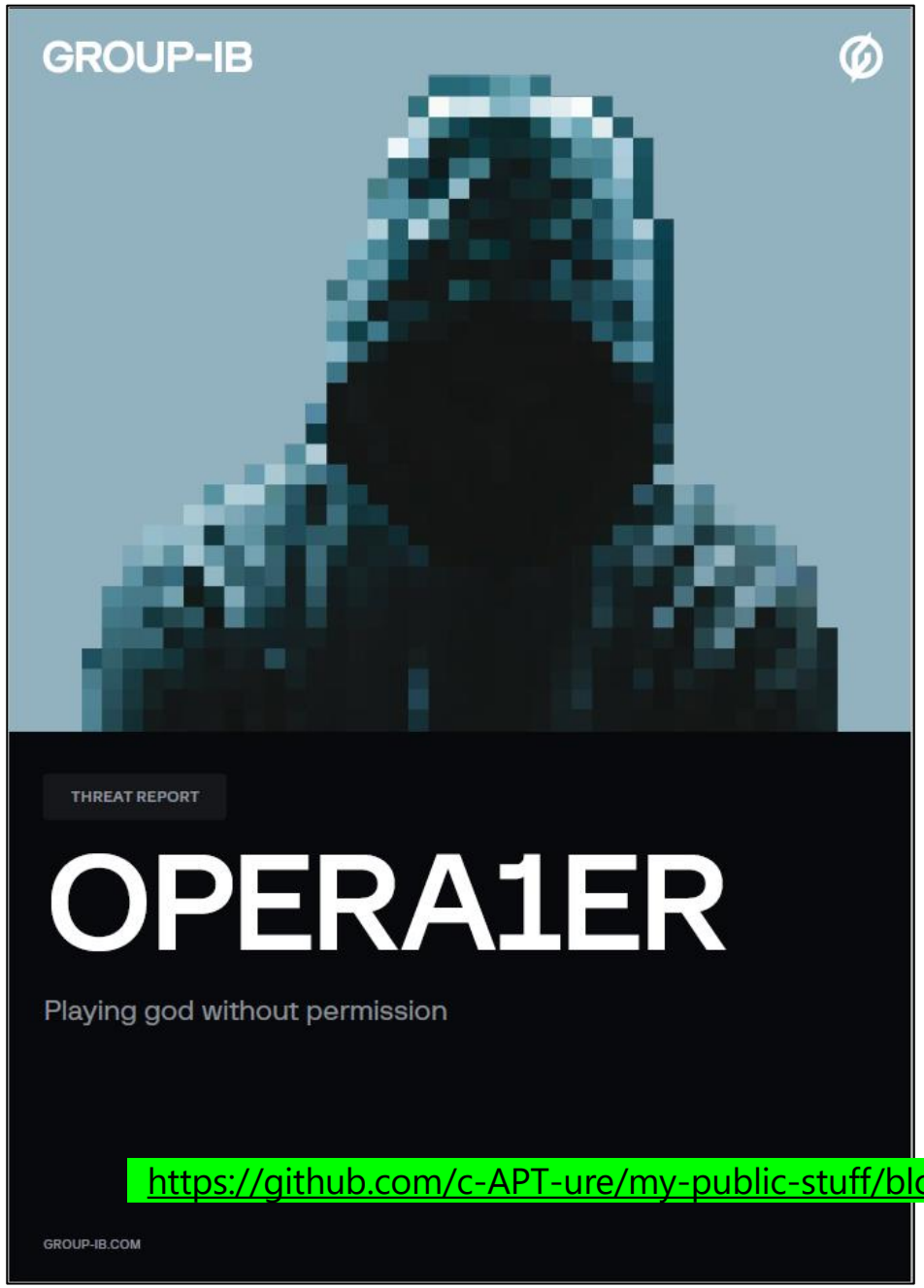
Who is "DESKTOP-Group"?

In 2019, I started sharing on Twitter about this TA, later starting to use the hashtag #DESKTOPgroup.



There is also a closed Google-group for research collaboration, mostly with people tracking or having access to emails or logs, related this TA's activity.

Malware samples and URLs have been shared and tagged on Abuse.ch Malware Bazaar or URLhaus.



https://github.com/c-APT-ure/my-public-stuff/blob/master/Group-IB_RPRT_OPERA1ER_EN_full.pdf

Timeline and Geography of attacks

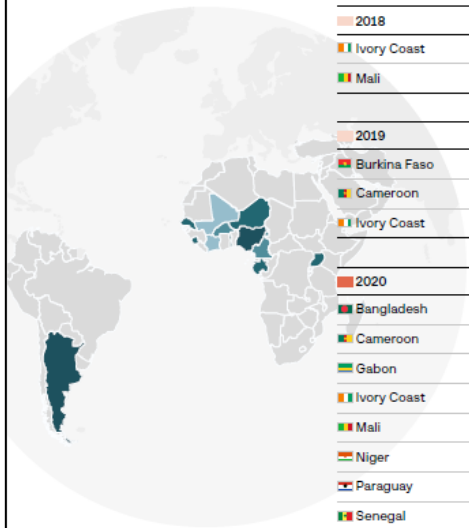
O for intrusions into IT systems
O for fraud operations

2018
2019
2020
2021

Findings from different investigation cases have been compiled here to provide a timeline of events that could be related to OPERA1ER's activity.

The cases depicted without the O symbol were obtained by Group-IB's passive monitoring of infrastructure. Not all warned organizations confirmed that they had money stolen from them. However, according to other sources the attacker successfully stole money via SWIFT (moderate confidence) and payment systems (high confidence).

Some incidents on the timeline actually belong to the same organization, as they were targeted more than once by the attacker. At least 15 different victims, whose infrastructure got hacked, have been identified to date.



2018	J	F	M	A	M	J	J	A	S	O	N	D	J
Ivory Coast					O							O	
Mali													O

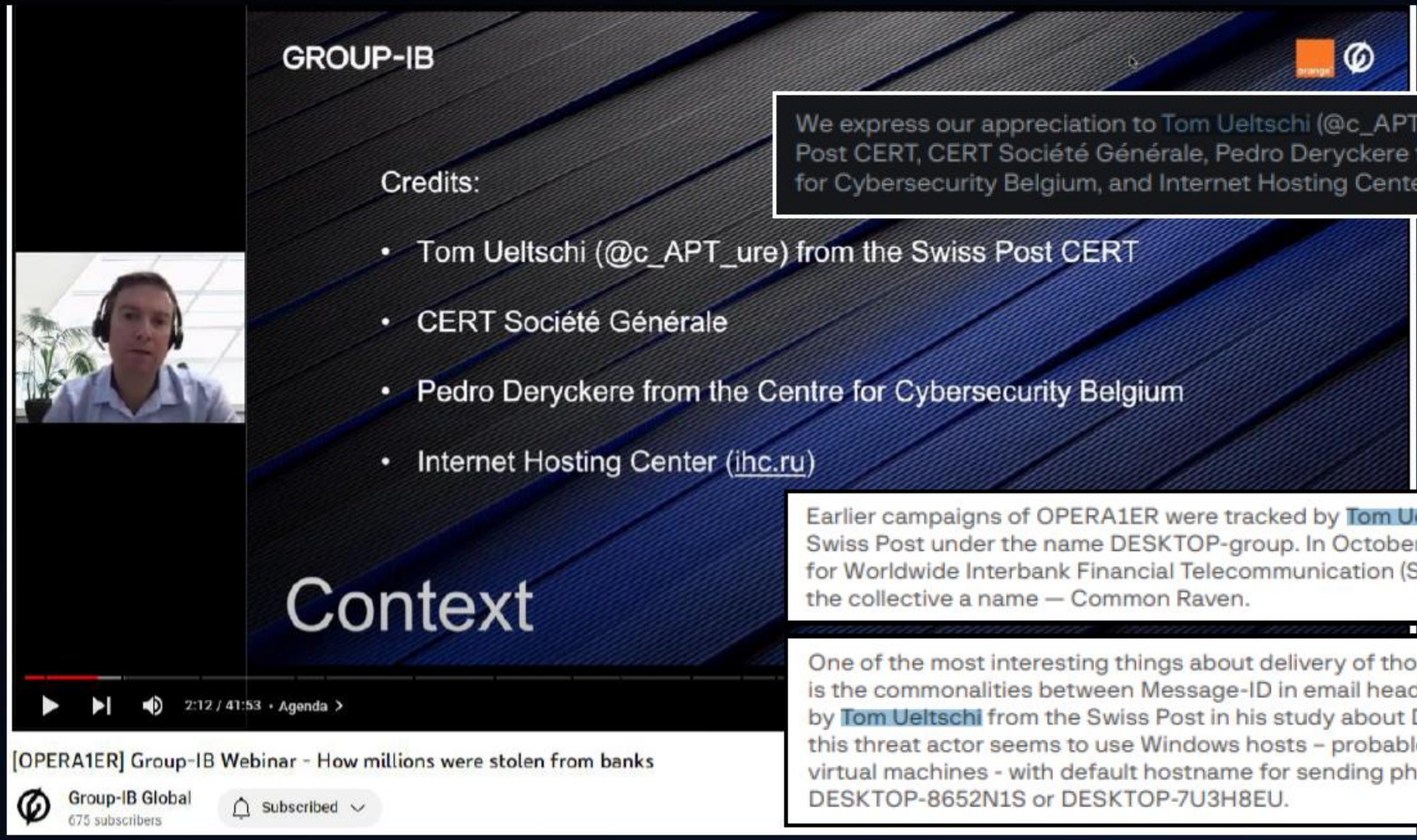
2019	J	F	M	A	M	J	J	A	S	O	N	D	J
Burkina Faso													O
Cameroon										O			
Ivory Coast		O			OO	OO	O				OO		

2020	J	F	M	A	M	J	J	A	S	O	N	D	J
Bangladesh					O								
Cameroon				O									
Gabon					O								
Ivory Coast				O	O								
Mali					O								
Niger								O					
Paraguay			O										
Senegal						OO							
Sierra Leone								O		O			
Togo					O								
Uganda										OO			

2021	J	F	M	A	M	J	J	A	S	O	N	D	J
Argentina				O									
Nigeria						O							

Thanks for Credits! (Group-IB & Orange Cyberdefense)

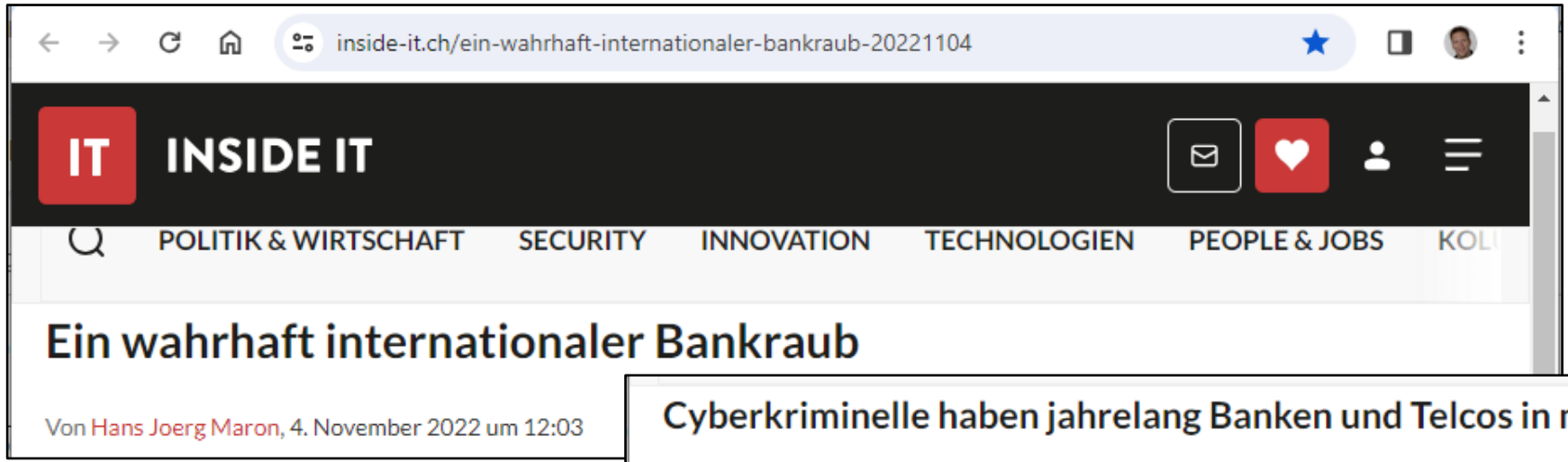




We express our appreciation to Tom Ueltschi (@c_APT_ure) from Swiss Post CERT, CERT Société Générale, Pedro Deryckere from the Centre for Cybersecurity Belgium, and Internet Hosting Center (ihc.ru).

Earlier campaigns of OPERA1ER were tracked by Tom Ueltschi from the Swiss Post under the name DESKTOP-group. In October 2020, the Society for Worldwide Interbank Financial Telecommunication (SWIFT) also gave the collective a name — Common Raven.

One of the most interesting things about delivery of those phishing emails is the commonalities between Message-ID in email headers. As noticed by Tom Ueltschi from the Swiss Post in his study about DESKTOPGroup, this threat actor seems to use Windows hosts – probable always the same virtual machines - with default hostname for sending phishing emails like: DESKTOP-8652N1S or DESKTOP-7U3H8EU.



Cyberkriminelle haben jahrelang Banken und Telcos in mindestens 15 Ländern beraubt.

Der französische Security-Dienstleister Group-IB hat einen detaillierten Bericht über das Vorgehen der Hacker-Gruppierung "Opera1er" veröffentlicht. Opera1er, auch bekannt als Desktop-Group, Common Raven oder NXSMS, hat demnach in den letzten rund 4 Jahren mehr als 30 Cyberangriffe auf Banken und Telcos in mindestens 15 Ländern durchgeführt. Dabei hat die anscheinend aus französischsprachigen Mitgliedern bestehende Bande mindestens 11 Millionen Dollar erbeutet. Laut Group-IB dürfte sie aber insgesamt über 30 Millionen Dollar gestohlen haben. Zwischen 2018 und 2022 gehörten Banken und andere Finanzinstitute in Argentinien, Paraguay, Bangladesch, Benin, Burkina Faso, der Elfenbeinküste, Gabun, Kamerun, Mali, Niger, Nigeria, Senegal, Sierra Leone, Uganda und Togo zu den Opfern.

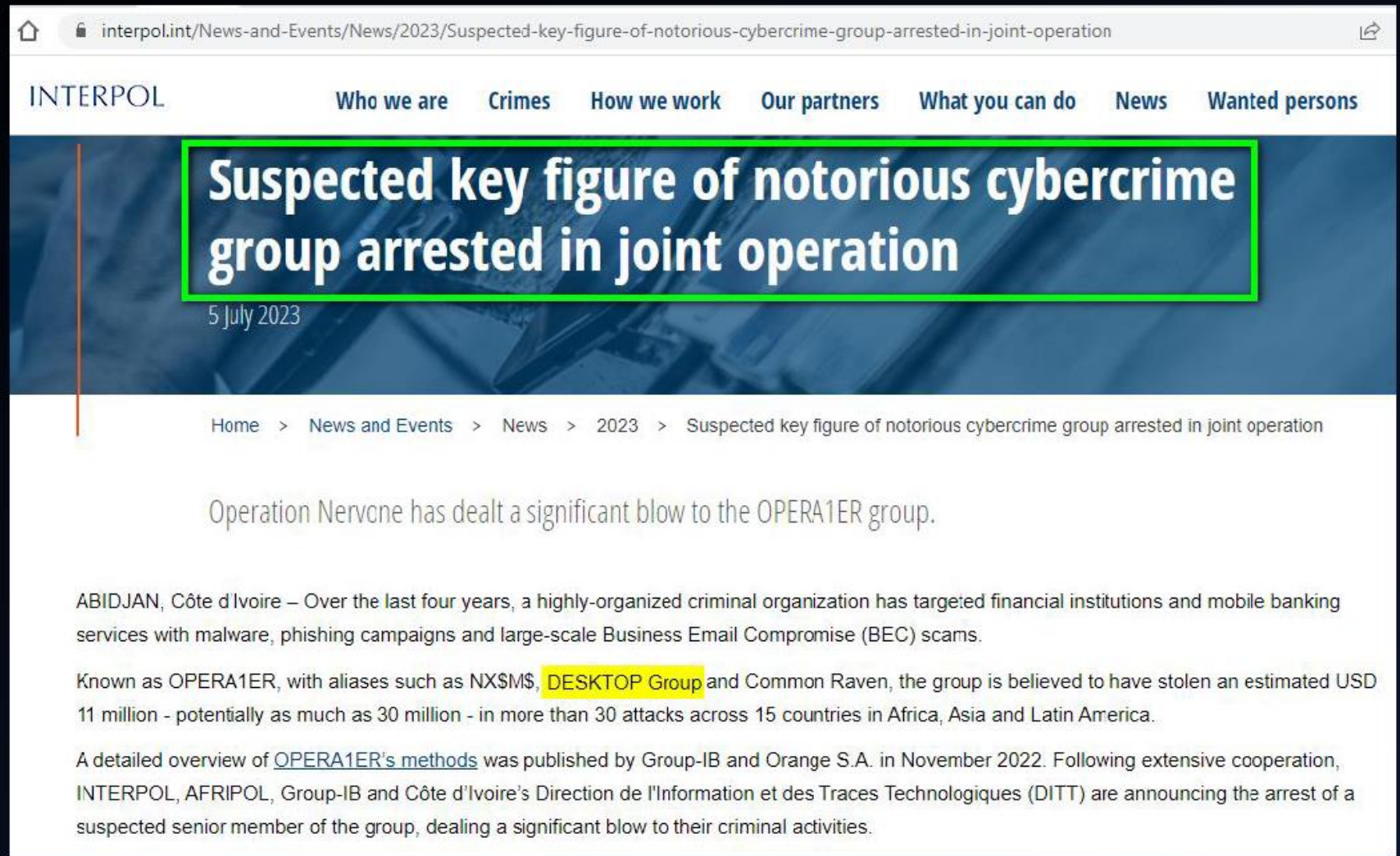
Die Aktivitäten von Opera1er wurden zuerst im Jahr 2019 vom CERT-Team von Orange bemerkt. Zu den Experten, die die Aktivitäten beobachteten, gehörte auch Tom Ueltschi vom CERT der schweizerischen Post, der von Group-IB mehrmals erwähnt wird. Ueltschi gab der Gruppe den Namen Desktop-Group.

Lange Spionagephase vor dem Raub

Laut dem aktuellen Bericht von Group-IB verwendet Opera1er keine fortschrittliche selbst entwickelte Software für ihre Angriffe, sondern Open-Source und andere Gratistools aus dem Darknet wie Cobalt Strike oder Metasploit.

Collaboration with INTERPOL

How it's going... (Jun – Jul 2023)



interpol.int/News-and-Events/News/2023/Suspected-key-figure-of-notorious-cybercrime-group-arrested-in-joint-operation

INTERPOL

Who we are | Crimes | How we work | Our partners | What you can do | News | Wanted persons

Suspected key figure of notorious cybercrime group arrested in joint operation

5 July 2023

Home > News and Events > News > 2023 > Suspected key figure of notorious cybercrime group arrested in joint operation

Operation Nervone has dealt a significant blow to the OPERA1ER group.

ABIDJAN, Côte d'Ivoire – Over the last four years, a highly-organized criminal organization has targeted financial institutions and mobile banking services with malware, phishing campaigns and large-scale Business Email Compromise (BEC) scams.

Known as OPERA1ER, with aliases such as NXM, DESKTOP Group and Common Raven, the group is believed to have stolen an estimated USD 11 million - potentially as much as 30 million - in more than 30 attacks across 15 countries in Africa, Asia and Latin America.

A detailed overview of OPERA1ER's methods was published by Group-IB and Orange S.A. in November 2022. Following extensive cooperation, INTERPOL, AFRIPOL, Group-IB and Côte d'Ivoire's Direction de l'Information et des Traces Technologiques (DITT) are announcing the arrest of a suspected senior member of the group, dealing a significant blow to their criminal activities.



How it happened

The group's illicit e-mail campaigns were first detected by Group-IB in 2018, when they recognized spear phishing operations responsible for spreading malware such as remote access tools.

Under the auspices of Operation Nervone, INTERPOL's Cybercrime Directorate, Group-IB, and third-party stakeholder Orange exchanged intelligence which helped track the group's behaviours and identify a probable location for their activities.

Additional information was provided by the United States Secret Service's Criminal Investigative Division and Booz Allen Hamilton DarkLabs cybersecurity researchers, confirming a number of leads.

In early June, authorities in Côte d'Ivoire were able to arrest a key suspect linked to attacks against financial institutions across Africa.

"Operation Nervone is a testament to what we can achieve through international collaboration and intelligence sharing. This successful operation marks a significant step in our ongoing mission to dismantle organized cybercrime networks, showcasing the power of collective action in stemming the tide against cybercrime."

Bernardo Pillot, INTERPOL's Assistant Director of Cybercrime Operations

Thanks!



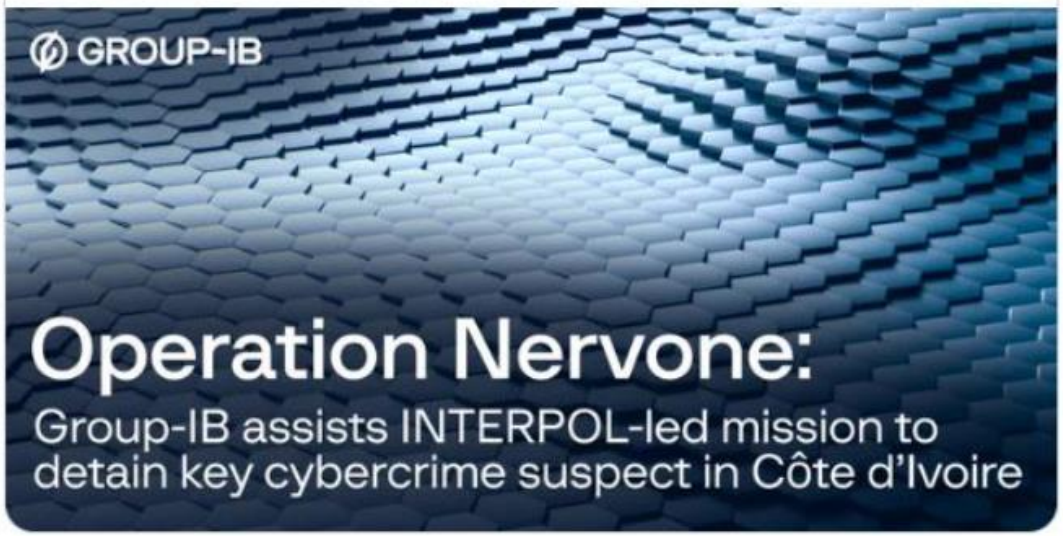
Rustam Mirkasymov
@Ta1ien Follows you

Threat Intel
Head of Cyber threat research dpt. @ Group-IB EU

Rustam Mirkasymov
@Ta1ien

Many thanks to everyone who contributed into the investigation and worked with us together (especially @OrangeCertCC and @c_APT_ure) in making the cyberspace safier.

> **Group-IB Global** @GroupIB · Jul 5
> Group-IB has assisted in the INTERPOL-led Operation Nervone, aimed at successfully disrupting the operations of the cybercriminal syndicate #OPERA1ER in French-speaking Africa. More details: bit.ly/3NHIU9j
>
> @INTERPOL_HQ @INTERPOL_Cyber



3:37 PM · Jul 5, 2023 · **2,306** Views

Is this the end?

Re-use of C&C domains

The following is just some short analysis, why I assess that DESKTOP-group is again and still active after the arrest of a key figure from their group via INTERPOL Operation Nervone.

So far, at least the following 3 DDNS domains have been re-used in attacks that were previously used by DESKTOP-group since 2020.

2020-10-01_23	bestsuccess.ddns[.]net	79[.]134.225.95	AveMaria / WarZone RAT
2020-10-10_137	bestsuccess.ddns[.]net	185[.]174.40.142	AveMaria / WarZone RAT
2020-10-29_123	bestsuccess.ddns[.]net	153[.]92.126.202	AveMaria / WarZone RAT
2020-12-02_27	bestsuccess.ddns[.]net	80[.]85.159.33	AveMaria / WarZone RAT
2021-10-19_67	bestsuccess.ddns[.]net	185[.]118.167.229	AveMaria / WarZone RAT
2021-11-18_49	bestsuccess.ddns[.]net	185[.]118.167.229	AveMaria / WarZone RAT
2022-07-30_717	bestsuccess.ddns[.]net	80[.]85.153.132	Remcos RAT
2022-08-15_133	bestsuccess.ddns[.]net	80[.]85.153.132	Remcos RAT
2022-11-16_106	bestsuccess.ddns[.]net	197[.]210.54.233	AveMaria / WarZone RAT
2022-11-24_61	nightmare4666.ddns[.]net	91[.]192.100.57	AveMaria / WarZone RAT
2022-11-29_21	nightmare4666.ddns[.]net	64[.]188.23.167	AveMaria / WarZone RAT
2023-04-14_20	nightmare4666.ddns[.]net	45[.]88.67.63	AveMaria / WarZone RAT
2023-05-02_9	nightmare4666.ddns[.]net	45[.]88.67.63	AveMaria / WarZone RAT
2023-05-04_8	donelpacino.ddns[.]net	212[.]193.30.230	AveMaria / WarZone RAT
2023-05-14_611	nightmare4666.ddns[.]net	212[.]193.30.230	AveMaria / WarZone RAT
2023-05-16_40	nightmare4666.ddns[.]net	212[.]193.30.230	AveMaria / WarZone RAT

** 2023-06-?? INTERPOL Operation Nervone arrest in early June **

*.ddns.net domains (VT: 1st seen)

- bestsuccess (2020-09-10)
- nightmare** (2022-11-21)
- donelpacino (2023-04-03)

AveMaria / WarZone RAT

Remcos RAT

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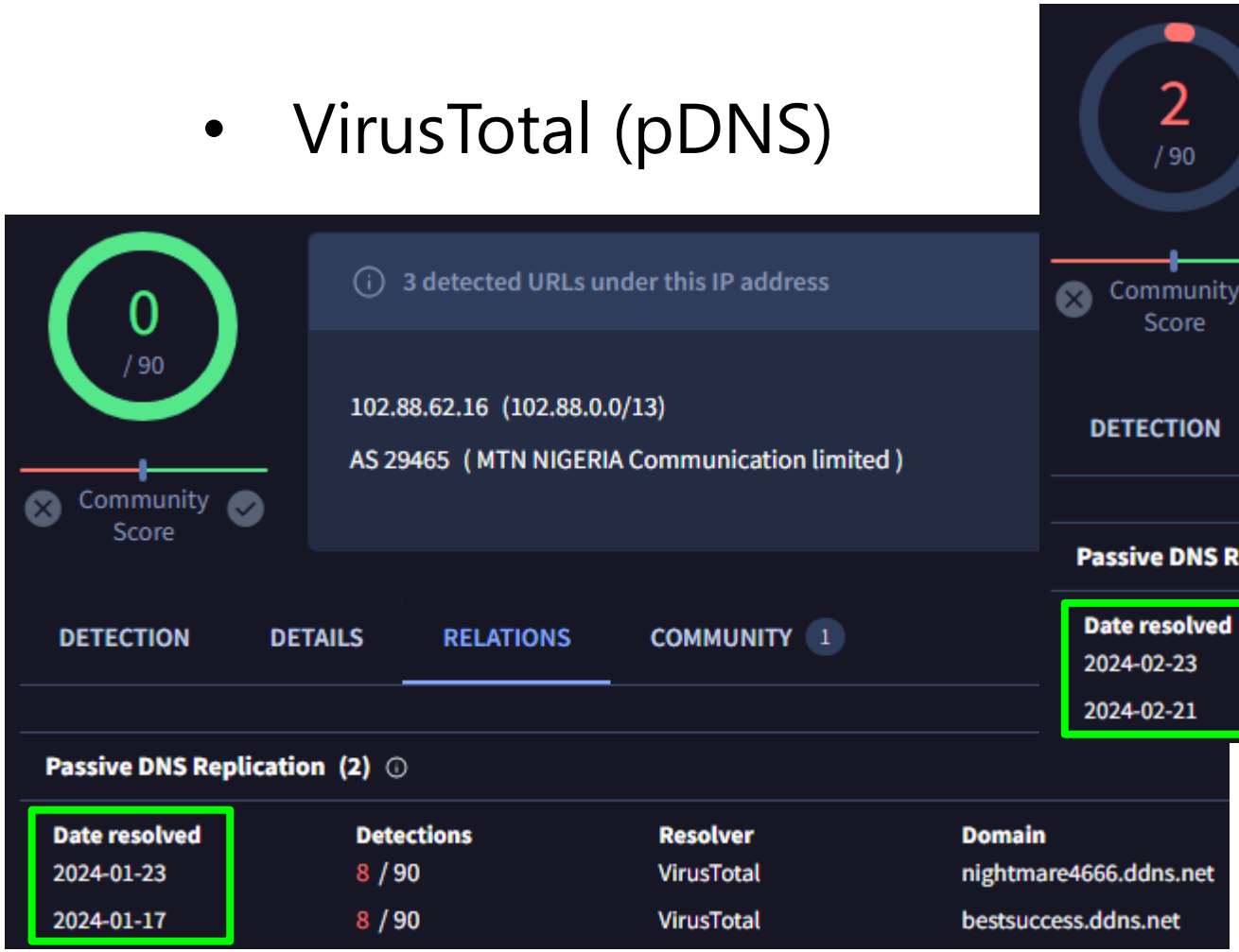
THE END ?? ☺

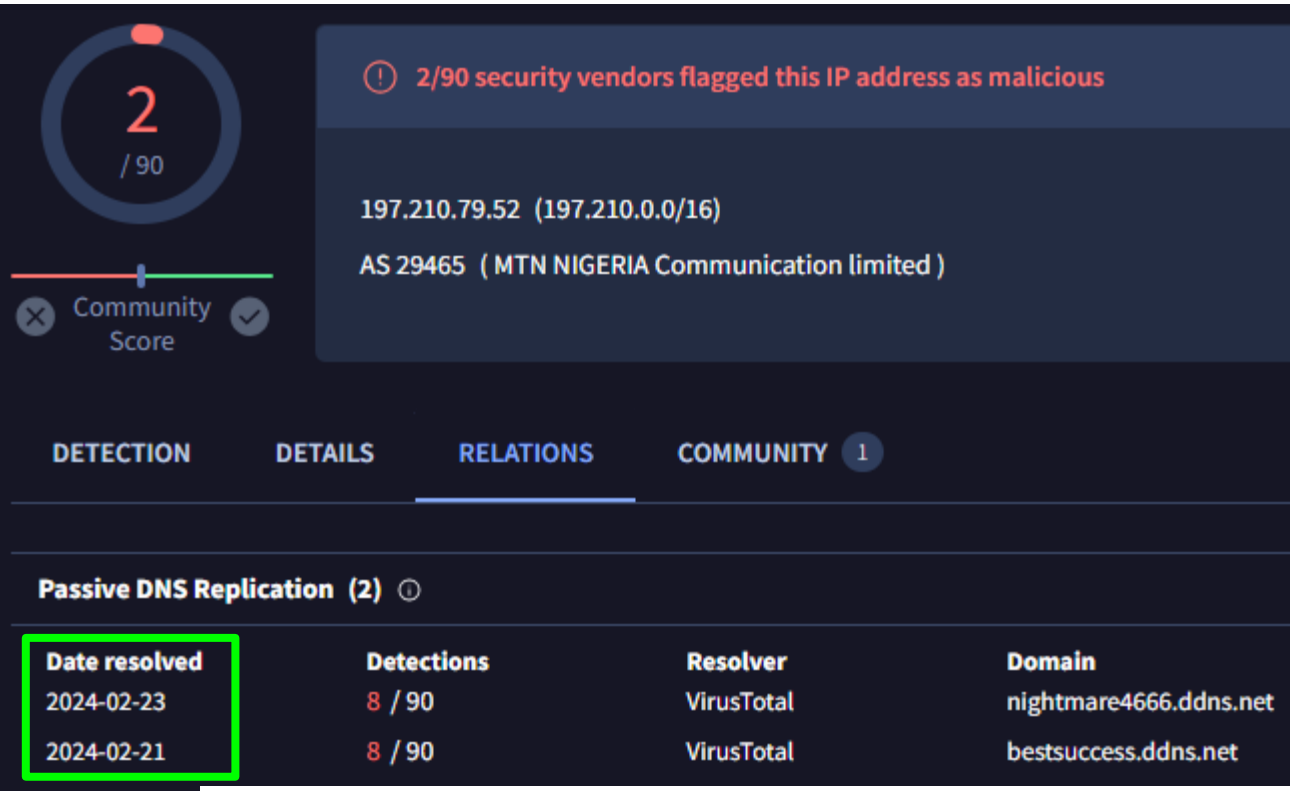
Re-use of C&C domains



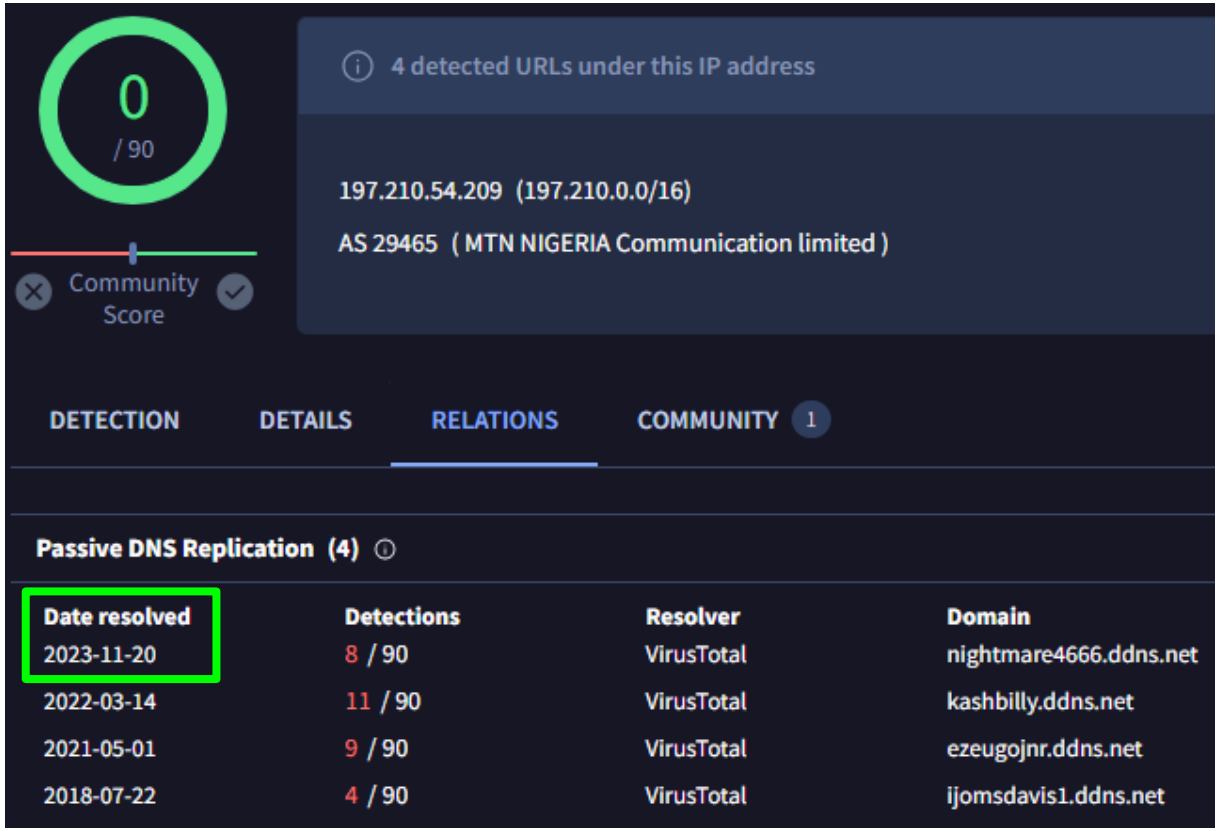
Re-use of C&C domains

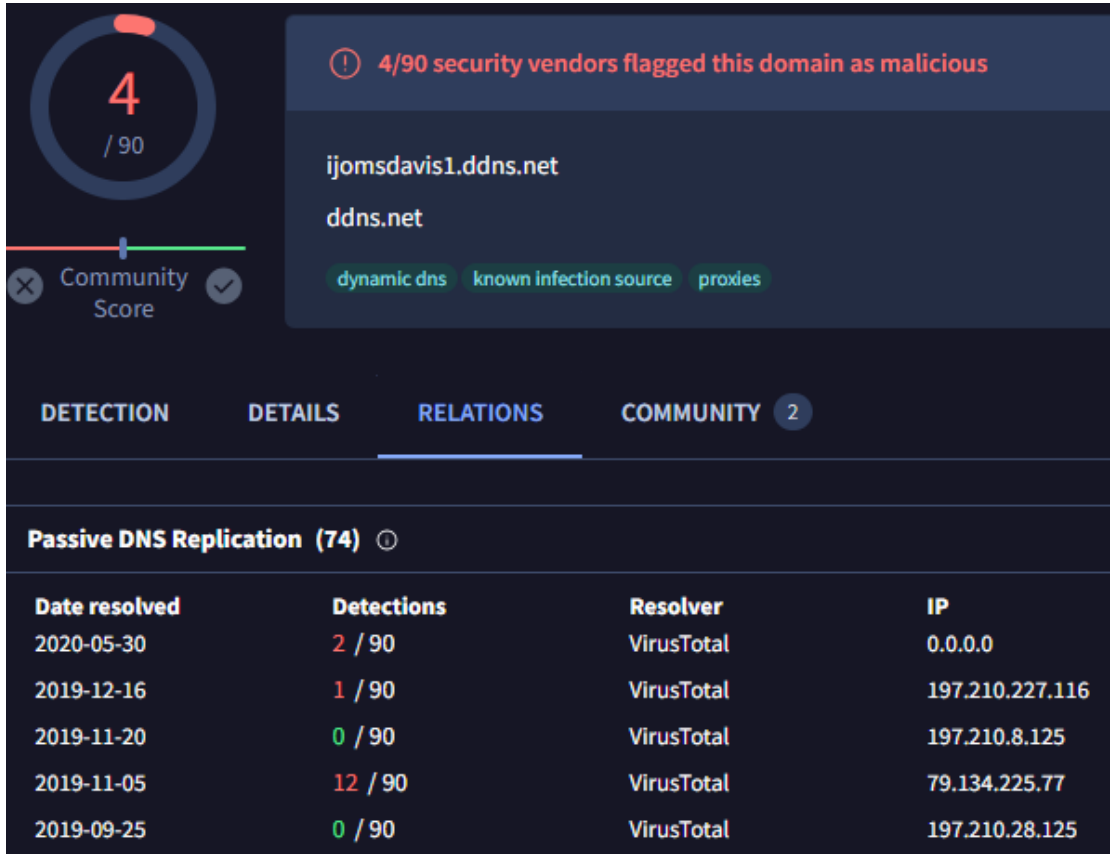
- VirusTotal (pDNS)



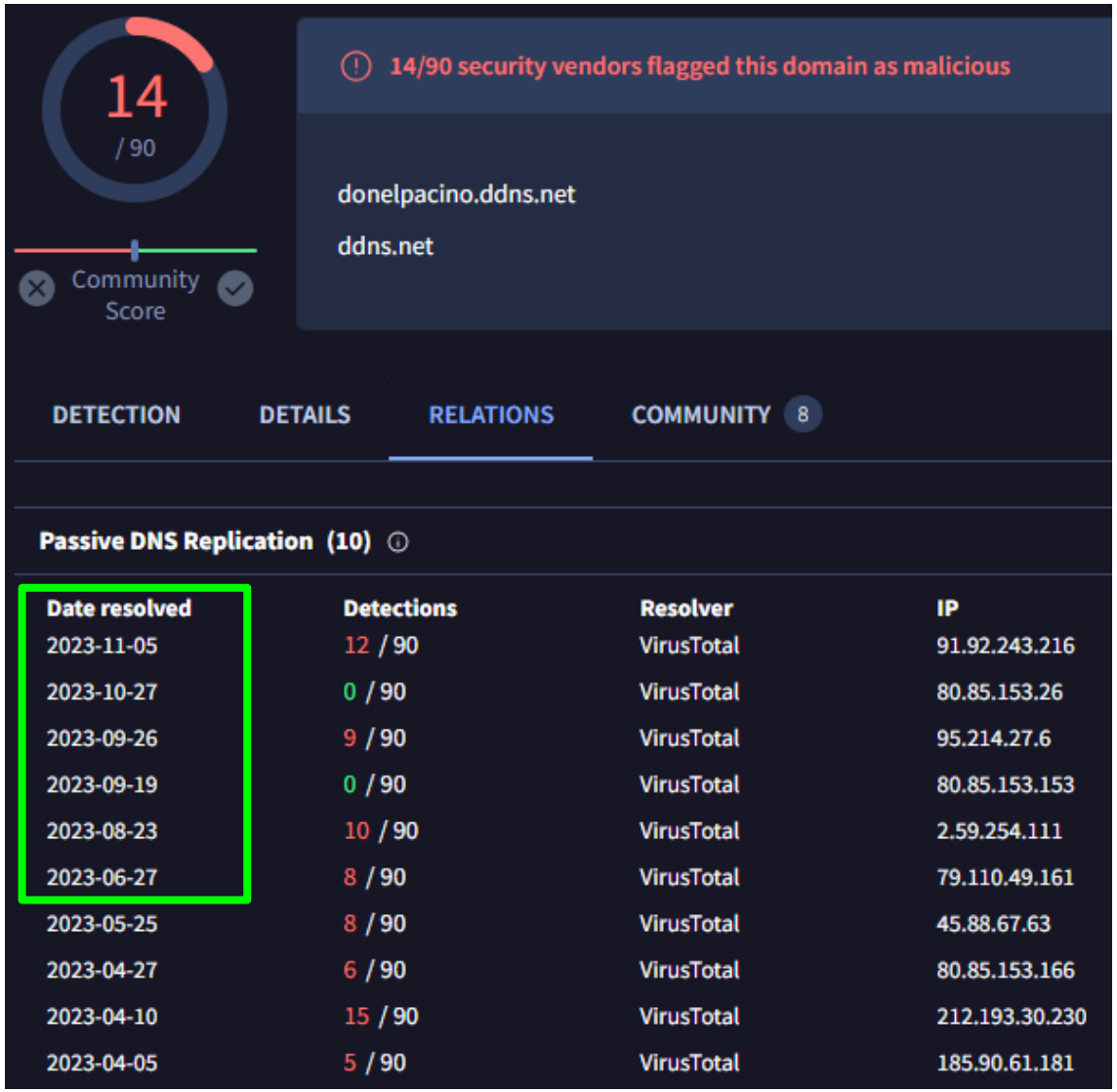


Re-use of C&C domains





Re-use of C&C domains



Thanks for your attention!
Questions?